

UPU-List of {a}

tid	u	rem	log P
1	5	3	-0.2
3	4	2	-0.5
6	2	1	-0.3

UPU-List of {e}

tid	u	rem	log P
1	2	1	-0.4
4	3	0	-0.6
6	1	0	-0.5

Join

 match tid

UPU-List of {a, e}

tid	u	rem	log P
1	7	1	-0.6
6	3	0	-0.8

On matching tid : $u = u_A + u_B$, $rem = \min(rem_A, rem_B)$, $\log P = \log P_A + \log P_B$

Aggregates: $EU = \sum u e^{\log P}$, $PTWU = \min(PTWU_A, PTWU_B)$, $PUB = \sum_{u+rem > 0} e^{\log P(u+rem)}$

$EP = 1 - \prod (1 - e^{\log P})$ accumulated in log-space